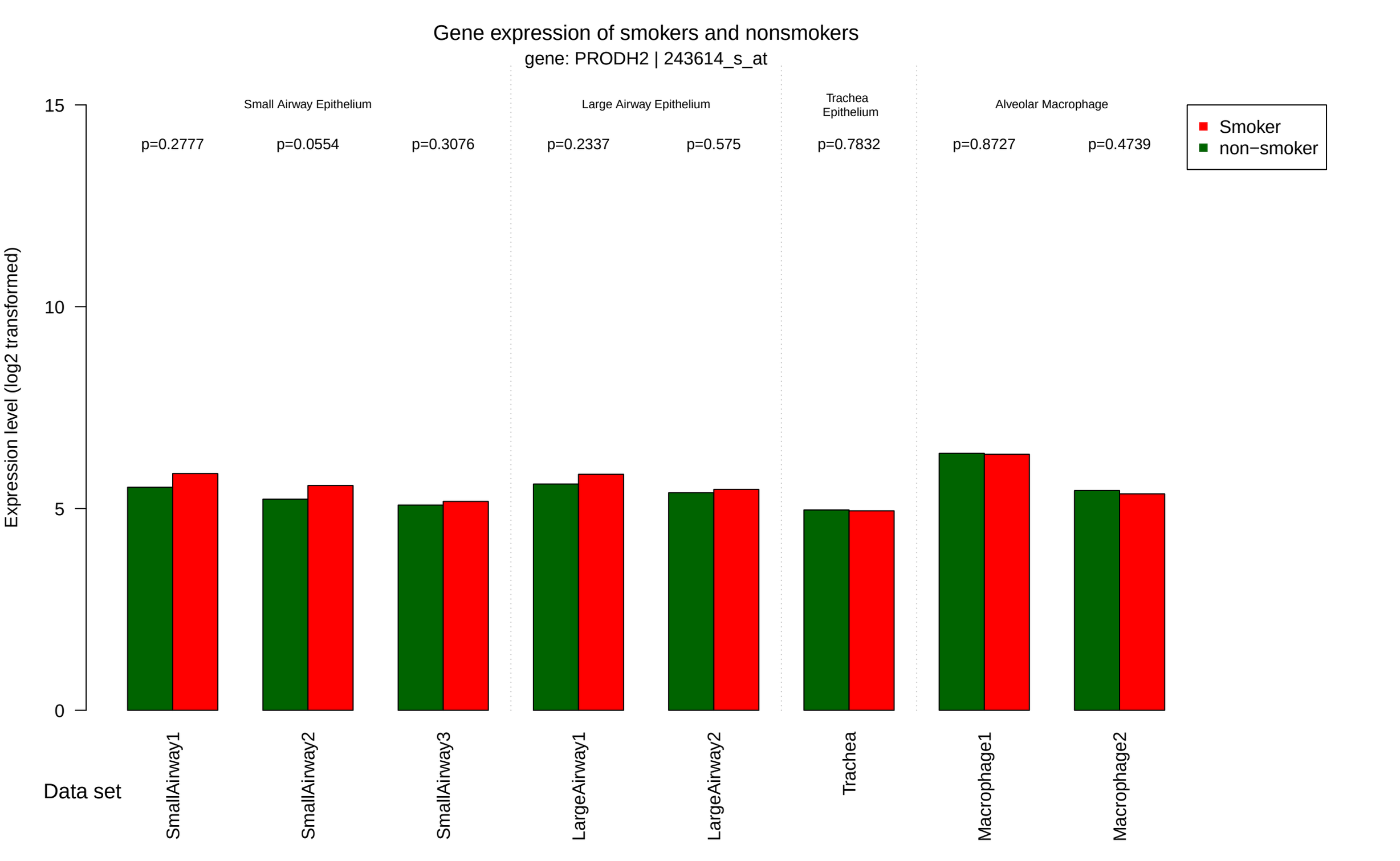
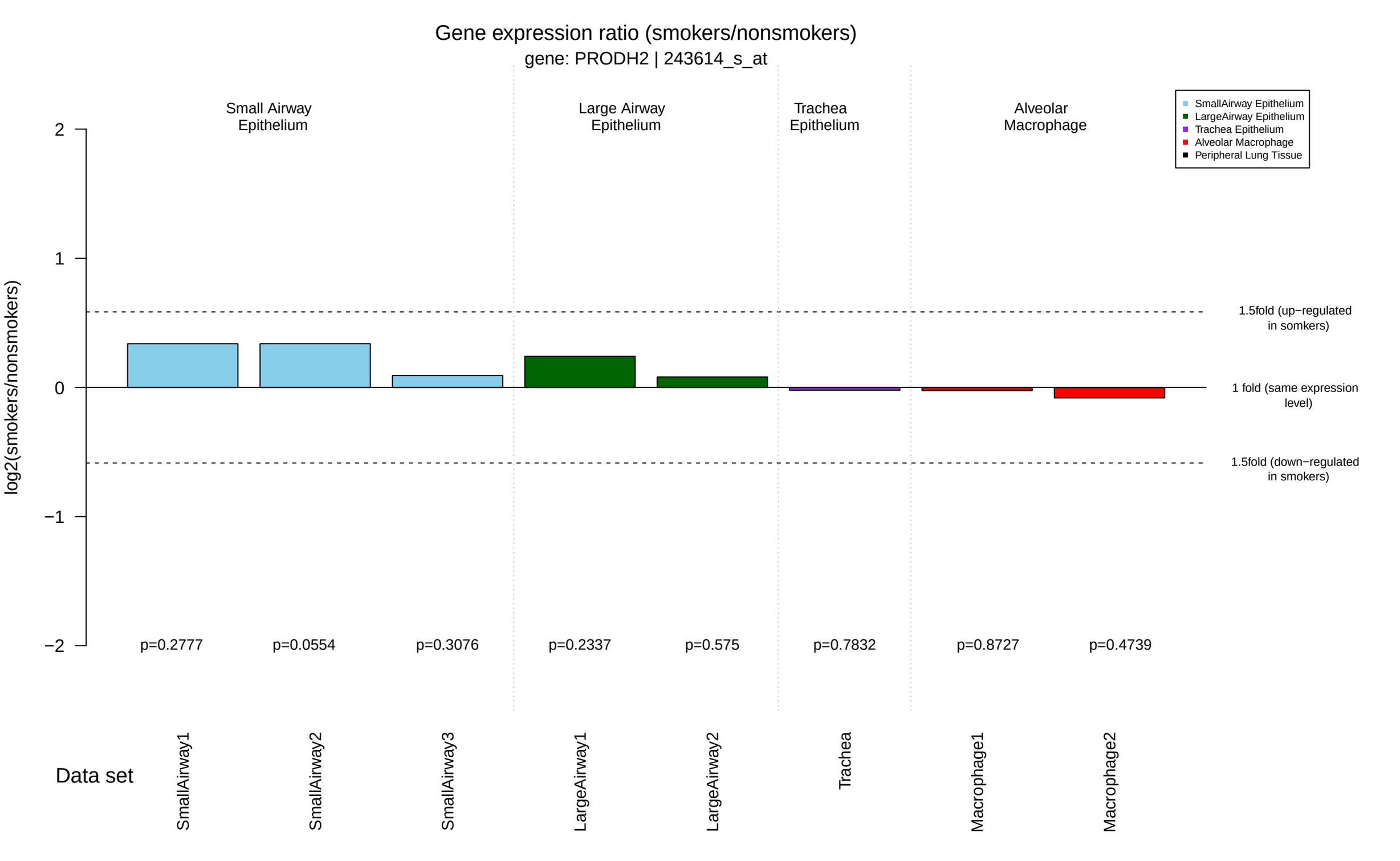
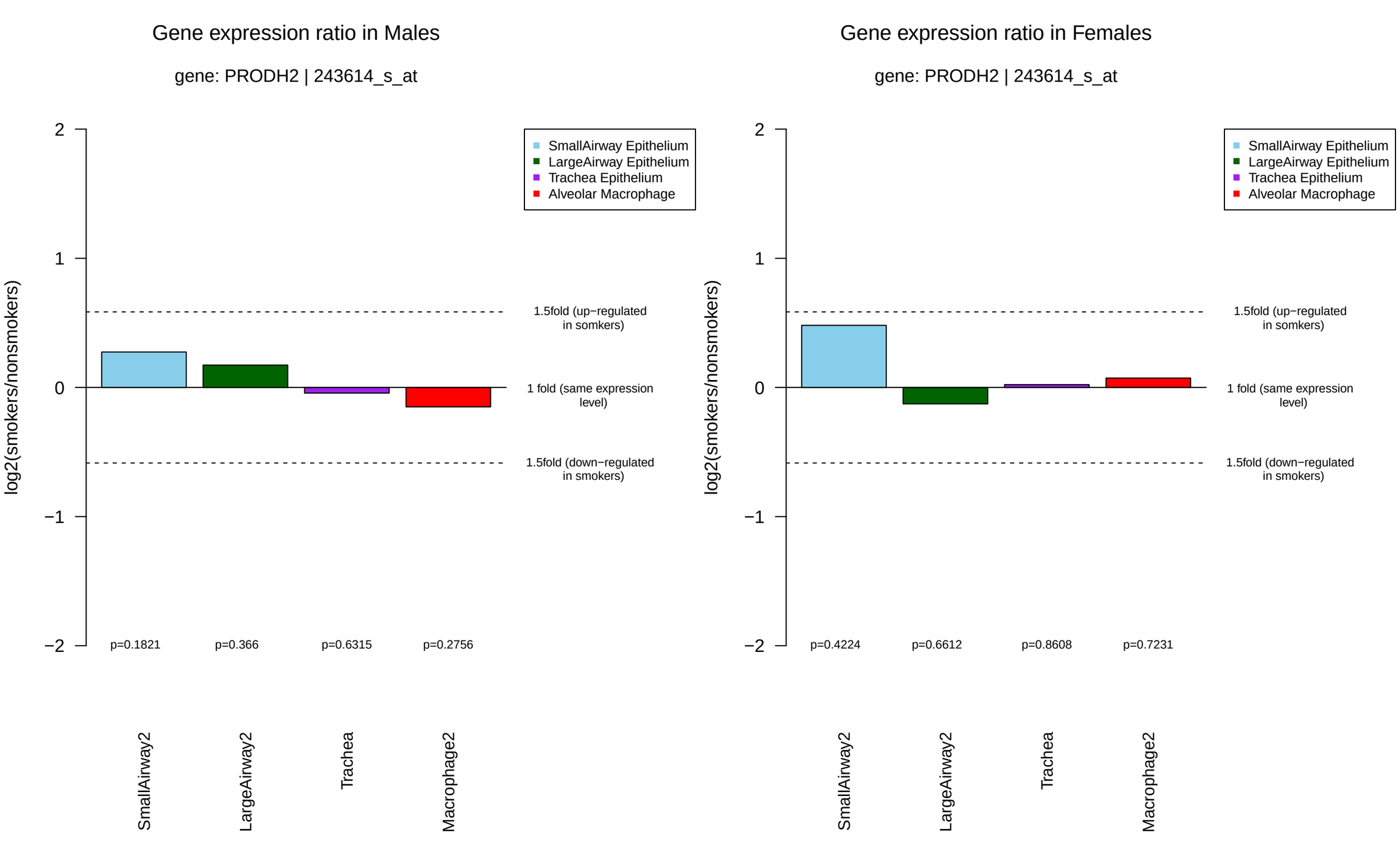
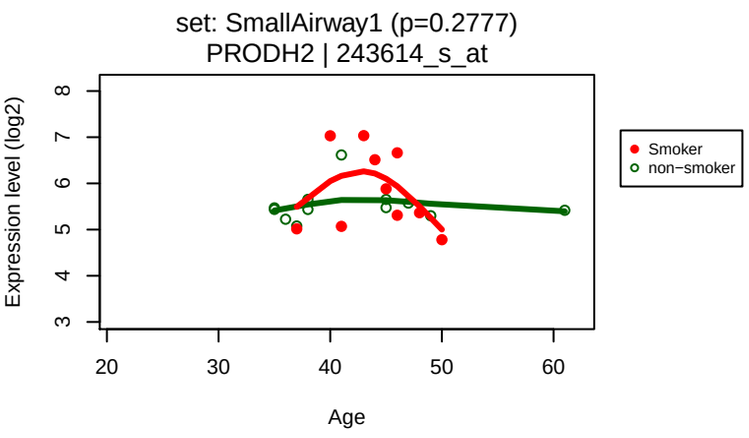
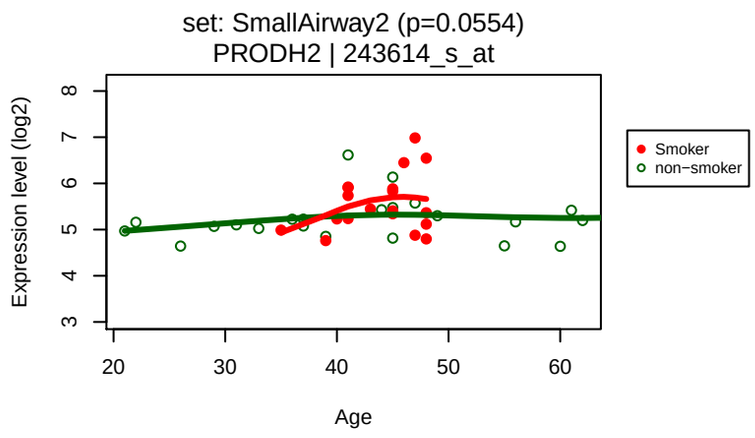
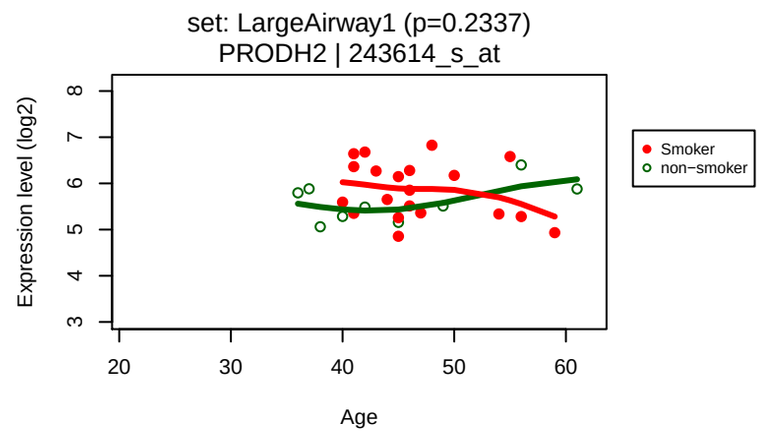

Gene expression of smokers and nonsmokers
gene: PRODH2 | 243614_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2777
p=0.0554
p=0.3076
p=0.2337
p=0.575
p=0.7832
p=0.8727
p=0.4739
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

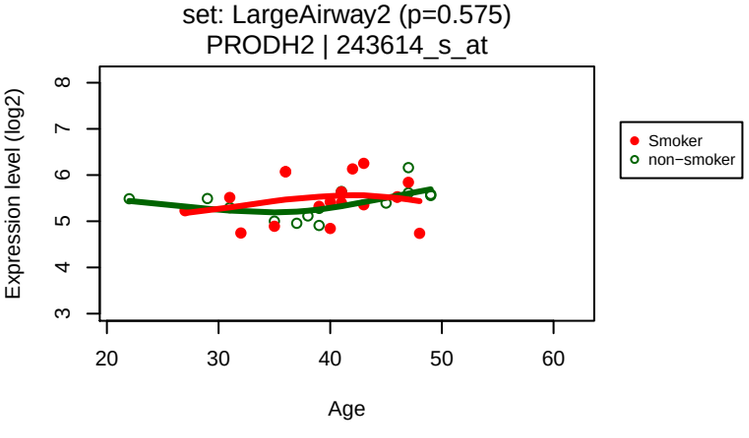

Gene expression ratio (smokers/nonsmokers)
gene: PRODH2 | 243614_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2777
p=0.0554
p=0.3076
p=0.2337
p=0.575
p=0.7832
p=0.8727
p=0.4739
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

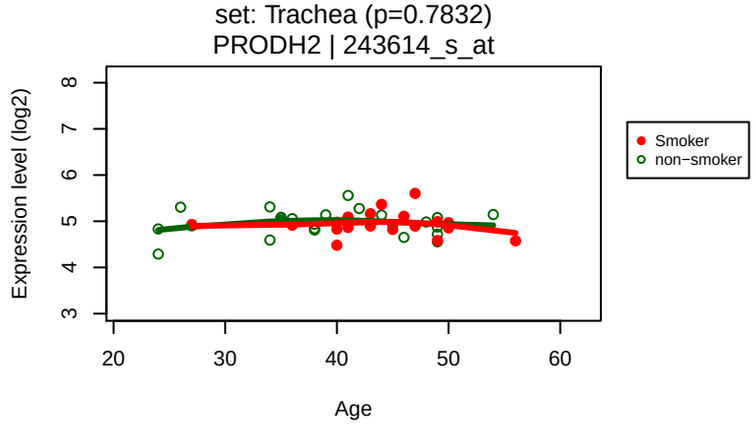

Gene expression ratio in Males
gene: PRODH2 | 243614_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1821
p=0.366
p=0.6315
p=0.2756
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRODH2 | 243614_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4224
p=0.6612
p=0.8608
p=0.7231
SmallAirway2
LargeAirway2
Trachea
Macrophage2

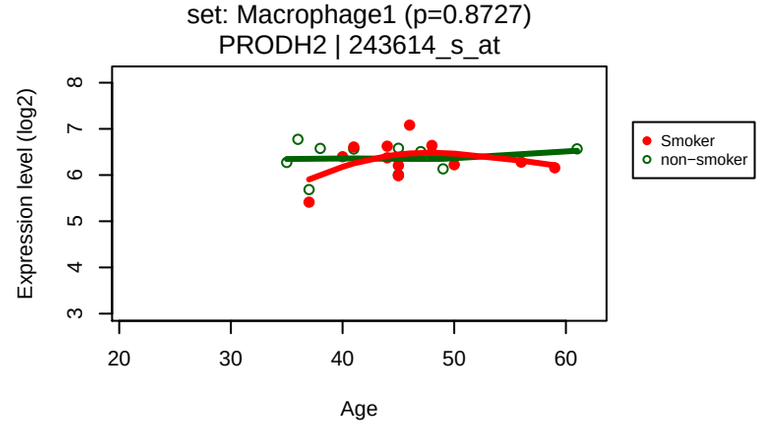

set: SmallAirway1 (p=0.2777)
PRODH2 | 243614_s_at
Expression level (log2)
Age
Smoker
non−smoker

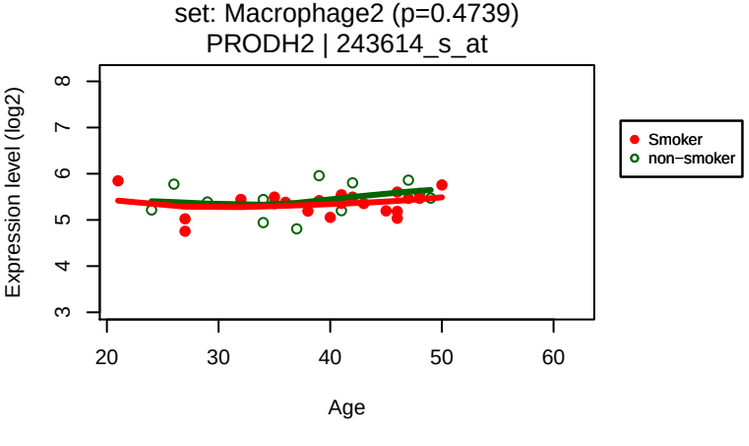

set: SmallAirway2 (p=0.0554)
PRODH2 | 243614_s_at
Expression level (log2)
Age
Smoker
non−smoker

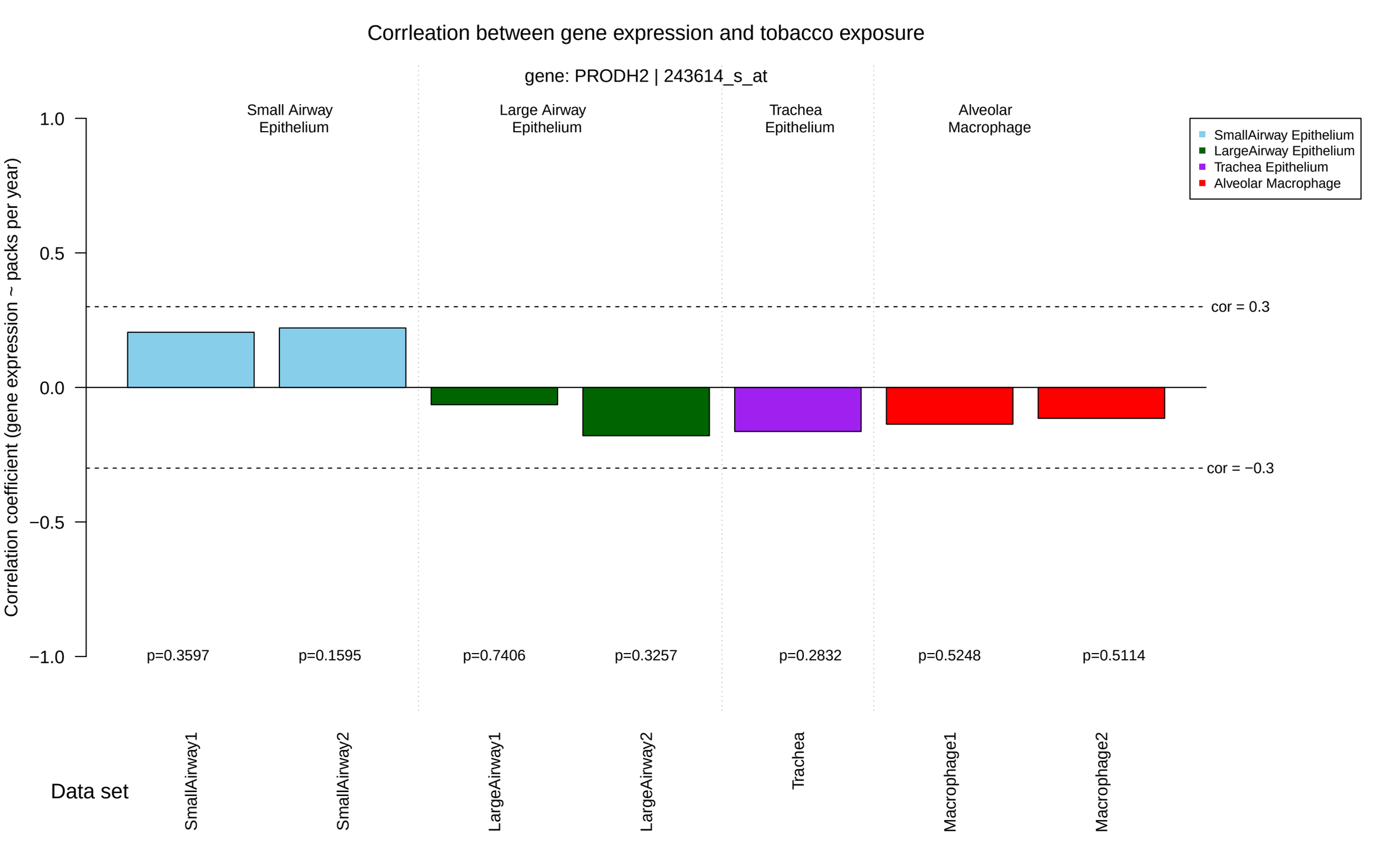

set: LargeAirway1 (p=0.2337)
PRODH2 | 243614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.575)
PRODH2 | 243614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7832)
PRODH2 | 243614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8727)
PRODH2 | 243614_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4739)
PRODH2 | 243614_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRODH2 | 243614_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3597
p=0.1595
p=0.7406
p=0.3257
p=0.2832
p=0.5248
p=0.5114
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2